DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: Electrical Engineering **Branch:** Electrical Engineering /Electrical Engineering (Electrical and Power)

Semester: 6 (VI)

Subject Code & Name: Industrial Automation and Control (BTEEE604A_Y19**)**

Max Marks: 60 **Date: 08/07/25** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	Automation Systems may include Control Systems but the reverse is not true.				CO1	1
	a. true	b. False	c. Not Applicable	d. All		
2	Automation also reduces ____________ of production significantly by efficient usage of energy, manpower and material.				CO1	1
	a. Limit	CO1	c. Oil	d. Quality		
3	The ___________ elements translate the physical process signals such as temperature, pressure or displacement to convenient electrical or pneumatic forms of information.				CO1	1
	a. Production	CO1	c. sensing	d. Generating		
4	Sensors and Actuators Layer is________________				CO1	1
	a. Level ‘0’	CO1	c. Level ‘2’	d. Level ‘3’		
5	A modern controller device used extensively for sequence control today in transfer lines, robotics, process control, and many other automated systems is Called __________				CO1	1
	a. Transfer	CO1	c. Sensor	d. Switching		
6	The PLC is essentially a microprocessor-based real-time computing system.				CO1	1
	a. True	CO1				

7	The digital inputs modules convert the external binary signals from the process to the internal ___________signal level of programmable controllers				CO1	1
	a. Analogue	CO1	c. Peak	d. Low		
8	Accuracy indicates the closeness of the measured value with the actual or true value.				CO1	1
	a. True	CO1				
9	Resistance thermometers employing metallic conductors for temperature measurement are called………………….				CO1	1
	a. MTD	CO1	c. Controllers	d. Scale		
10	One of below mentioned is not a type of Diaframgm.				CO1	1
	a. thin plate	CO1	c. Corrugated	d. thick Plate		
11	Potentiometers are displacement sensors. They can be used for linear as well as ……………………………. displacement measurement.				CO1	1
	a. Triangular	CO1	c. Temperature	d. Angular		
12	Which of the following is not an error in instrument readings				CO1	1
	a. Gross errors	b. Systematic errors	c. Temperature errors	d. Random Error		
Q. 2	**Solve the following.**					**12**
A)	**Describe the hierarchical structure of Industrial Automation System.**				CO2	**6**
B)	**Draw the block diagrammatic structure of – i) An Industrial Sensor.** **ii) An Industrial Actuator and explain in very short.**				CO2	**6**
Q.3	**Solve the following.**					**12**
A)	**Explain the construction and principle operation of bourdon tube pressure gauge.**				CO2	**6**
B)	**Describe Various methods of Temperature measurements.**				CO3	**6**
Q. 4	**Solve Any Two of the following.**					**12**
A)	**Explain Components of Hydraulic Actuation Systems.**				CO3	**6**
B)	**Describe the types of automation systems.**				CO3	**6**
C)	**Explain RTD with help of neat diagram, the variation of resistance of metals with temperature with suitable expression and draw the graphs for commonly used materials.**				CO3	**6**
Q.5	**Solve Any Two of the following.**					**12**

A)	**Explain Industrial Actuator Systems with suitable block diagram**	CO4	**6**
B)	**Explain Industrial Sensors and Instrument Systems with Proper Block Diagram.**	CO4	**6**
C)	**Draw Architecture of PLCs and explain What is a communication Processor?**	CO4	**6**
Q. 6	**Solve Any Two of the following.**		**12**
A)	**Explain with neat diagram electromagnetic type flow meter.**	CO5	**6**
B)	**Describe PID Controller in detail.**	CO5	**6**
C)	**Explain Components of Hydraulic Actuation Systems.**	CO5	**6**
	***** End *****		